

VAST Challenge 2026 题目原文+翻译

Mini-Challenge 1

Background

原文

TenantThread is a property tech company that sells property management software to landlords and multifamily operators. On the surface, its tools cover workflow and maintenance ticketing. But one of its core products, the “Retention Optimizer”, scores individual tenants based on their payment timing, maintenance requests, and even the tone of their messages, then uses those scores to flag residents as “high-friction” or recommend renewal pressure tactics. The company has publicly claimed the data is anonymized, but internal documents show that in smaller properties, cross-referencing a unit type with a payment date is enough to put a name to a score. A local newspaper, the SaltWind Journal, has been running an ongoing exposé on these practices, and the hashtag "AlgorithmicEviction" has been gaining traction on social media for weeks.

In the middle of all this, TenantThread quietly signed a merger agreement with CivicLoom Realty Partners — a major real estate owner-operator. The deal, codenamed “Project HarborCrest,” would rebrand TenantThread’s platform and bring in significant capital. The merger announcement is under a strict information embargo until 6:00 PM on June 5th, 2046. An automated compliance tool, referred to as “The Judge”, was installed to ensure information was not leaked. But around 5:00 PM that afternoon, embargoed information about the deal started showing up on FleX (a social media service), by accounts associated with TenantThread’s automated communications and public reactions system. The embargo was broken!

CivicLoom’s legal team now needs to understand what happened. You are assisting that team. Your job is to dig into TenantThread’s internal communications from the two weeks leading up to the breach and reconstruct what was going. You have information from AI agents that managed TenantThread’s corporate communications. This includes the messages, internal deliberations, and public post. Build a visual analytics system to help determine: Did TenantThread’s team deliberately leak the merger or did the system simply break down under pressure?

翻译

TenantThread是一家房产科技公司，主要向房东以及多户住宅运营方售卖物业管理软件。表面上，该公司的工具仅用于处理工作流程与维修工单业务，但其核心产品之一“租户留存优化系统”，会依据租户缴

费时效、报修申请情况甚至聊天消息语气对租户进行打分，再根据评分将住户标记为“高矛盾租户”，或是推送租房续约施压手段。该公司对外宣称用户数据已完成匿名处理，但内部文件显示，在小型楼盘中，仅结合户型与缴费日期就能对应到具体租户身份。地方媒体《盐风日报》一直在持续曝光这类违规行为，“算法驱逐租户”相关话题也已在社交平台持续走红数周。

在此背景下，TenantThread悄然与大型房产运营企业CivicLoom Realty Partners签订并购协议。这项代号为“海港之巅计划”的合作，将完成TenantThread平台品牌升级并引入大额资金。此次并购消息实行严格保密管控，官宣时间定为2046年6月5日下午6点，企业还部署了名为“审判者”的自动化合规工具防止信息外泄。但在当日下午5点左右，本该保密的并购信息，经由TenantThread自动化沟通与舆情响应系统关联账号，泄露至社交平台FleX，保密禁令彻底失效。

CivicLoom法务团队急需查清事件原委，本次任务由你协助完成。你需要梳理泄密事件发生前两周内TenantThread的全部内部沟通内容，还原事件全貌。你可调取该企业负责企业对外对内沟通的智能代理相关数据，包含聊天记录、内部商议内容以及对外公开动态。你需要搭建一套可视分析系统，最终判定此次并购消息泄露，是团队人为故意泄露，还是系统运行异常意外导致。

Tasks and Questions

原文

1. What were the key events and relationships that led up to the inappropriate release? Visualize the sequence of events. Include elements such as:
 - Key actions
 - Causal relationships
 - Decision points and actors involved in those decisions
 - Highlight decisions and system elements that were important to the post making it past the embargo enforcement
2. The evasion of the embargo was a new behavior. Provide visual analytics that discover and illustrate the typical behavior of systems/agents/people involved. How does the behavior that led to the release compare to the prior behaviors?
3. Were there leading indicators that such a release was possible?
 - Are there prior occasions where agent's actual behaviors differ from their expected behavior? When? Which behaviors were observed?
 - Were there other occasions where the agent system exhibited behaviors like those that led to the release? When? Which behaviors were observed?
 - Why didn't the prior occasions result in noticeable action?

翻译

1. 梳理造成机密信息违规泄露的核心事件与关联关系，可视化还原完整事件时序，内容需涵盖：
 - 核心操作行为
 - 事件因果关系
 - 决策节点及参与决策人员
 - 重点标注出绕过保密管控、最终造成信息外泄的关键决策与系统相关要素。
2. 此次违规泄露属于全新异常行为，请通过可视分析手段挖掘并展示相关人员、智能代理、业务系统的日常常规运行模式，对比本次泄密相关行为与以往常规行为存在的差异。
3. 事件发生前是否存在能够预判信息泄露风险的预警征兆？
 - 过往是否出现过智能代理实际运行行为偏离预设规范的情况？若有，请说明发生时间以及具体异常行为。
 - 智能代理系统此前是否出现过与本次泄密成因相似的异常行为？若有，请说明发生时间以及具体行为表现。
 - 过往同类异常行为为何没有引发正式管控处理与风险应对？

Mini-Challenge 2

Background

原文

You work in the AI systems administration department of Company A. A post was recently made on SaidIT looks like a malfunction of an agent: It is in gibberish and in a forum that appears random. Your job is to track down why this post occurred and suggest a remedy to the situation.

The anomalous posting was sent to SaidIt by John Windward on May 17, 2046 at 4:21am.

The agentic system is complex and has many automated behaviors in it. Nobody has taken the time to map out the whole system. Nobody knows all of the interactions that are possible or are being used. The overall goal is to understand what is actually present. As with any complex system, the first step is to map out the activity and develop a baseline, then compare that to the specific system behavior of interest and finally to suggest what might be done to prevent future issues.

Build a visual analytics system that integrates visuals with algorithms to enable interactive exploration and documentation of the multi-agent system!

翻译

你任职于A公司人工智能系统运维部门，近期社交平台SaidIT出现一条异常动态，内容杂乱无意义，发布板块也毫无规律，初步判定为智能代理运行故障所致。你的任务是查清该异常动态产生的根本原因，并给出对应的整改解决办法。

该异常内容由约翰·温沃德账号于2046年5月17日凌晨4点21分发布在SaidIT平台。

企业智能代理系统架构繁杂，内置大量自动化运行逻辑，目前尚无人员梳理出完整的系统架构，各类潜在交互模式与运行机制也未被全部掌握。本次任务核心目标是摸清系统真实运行状态。排查复杂系统问题需遵循固定流程：首先梳理系统全部运行行为、建立正常运行基准模型，再对标分析目标异常行为，最后制定可行方案避免同类问题再次发生。

请搭建一套融合可视化展示与数据分析算法的可视分析系统，实现对多智能代理系统的交互式数据探查与行为溯源记录。

Tasks and Questions

原文

1. How was the anomalous SaidIT posts made? Build a visual tool that describes the sequence of events that led up to the post. Include people and system interactions that were important to the post being made.
 - Provide a detailed view that focuses on the exact chain of events for this message of interest.
 - Provide a system overview that contextualizes this message chain in that overall system.
2. What do the posts “mean”? What is the origin of their contents? Illustrate the reasoning behind your assertion of their meaning as well as the probable content.
3. There are concerns that this behavior could repeat!
 - Investigate historic system behavior to find other instances of the behavior. Are there prior issues? Illustrate these prior occurrences and contrast them with the most recent one.
 - Illustrate possible system changes that would make it more difficult to occur in the future. Pick at most one spots to make an intervention. Using the data provided and visual analytics, illustrate why this location would likely be effective.

翻译

1. 查清SaidIT平台异常动态的完整生成流程，搭建可视化工具还原事件全过程，梳理促成此次异常发布行为的人员操作与系统交互关键环节。

- 精细化梳理目标异常信息从生成到发布的完整行为链路并进行可视化展示。
 - 绘制整体系统架构总图，明确此次异常行为链路在整个智能系统中所处的位置与关联模块。
2. 解读异常动态文本的实际含义，追溯文本内容来源，清晰说明含义判定依据以及合理的内容推断结果。
 3. 团队担忧此类系统异常行为会再次发生。
 - 调取系统历史运行数据，筛查同类异常事件，确认是否存在过往同类故障，展示历史异常事件并与本次最新异常事件进行对比分析。
 - 提出能够降低同类故障发生概率的系统优化调整方案，**最多仅选择一处核心节点进行优化干预**，结合赛事提供数据与可视分析结果，论证该优化节点具备实际防控效果的原因。

VAST Challenge 2026 Submission Instructions

Deadline

原文

The submission deadline is July 17, 2026 at 11:59 pm AOE.

Participants are permitted to submit multiple entries, but each entry must contain the complete set of required materials as specified below.

Note on GitHub Entries

In recent years there has been an increase in submissions through github including links to live demonstrations. This is great! However, please submit your entry with a github tag or as a release. In addition, entry forms must be submitted in full prior to the deadline. It is not permissible to provide a link to an entry form in github. Entries cannot be updated once submitted unless specifically requested by the VAST Challenge committee.

翻译

提交截止时间为2026年7月17日世界最晚时区晚间23点59分。

参赛队伍可提交多份参赛作品，且每份作品都必须配齐下文规定的全部提交材料。

GitHub提交相关说明：近年来越来越多参赛者通过GitHub提交作品并附上在线演示链接，该提交方式予以认可。提交时请为GitHub项目添加版本标签或发布正式版本，所有参赛表格务必在截止日期前完整提

交，禁止仅在GitHub放置报名表链接。作品提交完成后不得擅自修改，仅在赛事组委会正式许可后才可进行调整。

Required Materials

原文

Your completed submission package is due on July 17, 2026. Your submission package consists of a completed summary form and a four-minute video showing your visual analytics software in operation.

Please complete the summary form as follows:

Rename the summary form file to “index.htm”. Make sure to leave the form in “.htm” format. Use Microsoft Word or other word processor to edit the form.

Name your entry using a composite of your team’s organization, primary contact’s last name, and the challenge you are entering. For example, for a submission from the University of Maryland for Mini-Challenge 2, from a team led by Dr. Jones, please use UMD-Jones-MC2. If you are submitting multiple entries, please add a number to the end of the file name (e.g. UMD-Jones-MC2-1).

Provide a list of team members, their affiliation, and email addresses. Designate one team member as the Primary Contact. The primary contact must be able to respond to questions and communications related to the submission.

Indicate if this is a student team. A student team is defined as one led by a student and worked on by students. Class projects are good examples of submissions that would be provided by student teams.

Provide a list of analytic tools used in developing your solution. Commercially available tools like Tableau or SAS can be mentioned by name. For tools not commercially available, please provide a tool name, developer name (or a company name), and any links to the tool that can provide us more information about it. For tools developed by the submitting team, additional information such as where it was developed (e.g., “SPINVIZ was developed by the University of West Birmingham CS 459 Information Visualization class, taught Spring 2011 by Dr. Smith, and adapted by the student team for the challenge.”)

Provide an estimate of the number of hours spent by the entire team working on this submission. If you are submitting multiple entries, this number should reflect the time spent on this specific entry. Indicate whether you give your permission for your submission to be posted in the publicly accessible Visual Analytics Benchmark Repository.

Provide your explanatory video with voice narration. Participants are required to include a video of no more than four minutes in length. This video should be in .wmv format. If size permits, include this video in your electronic submission package. If not, post it on the internet and provide a link. Verify the link is active throughout the contest period, as it may be downloaded at various times by different VAST Challenge Committee members and reviewers. If you do not have a site on which you can post

your video, please contact vast_challenge@ieevis.org to discuss other approaches. You will also be asked to provide an electronic copy of the video for the archive repository. The following tips may be helpful in creating your video:

Provide a clear explanation how the visual analytics software you developed or used helped you analyze the data.

Give a clear explanation of your starting point for analysis and how you selected it

Explain your rationale for important design decisions.

Explain how the visualization and interactions helped in the analysis.

If filters or transformations are applied, explain what was used and in what sequence they were applied.

If there are anomalies shown in a visualization, explain how they are factored into the analysis.

Give a careful definition of terms used such as “uncertainty.”

Describe any important assumptions you made.

Answer the specific questions for this mini-challenge. These questions appear in the submission form.

翻译

全套参赛材料需在2026年7月17日前提交完毕，材料包含填写完整的项目总结表、时长4分钟以内的可视分析软件实操演示视频。

项目总结表填写规范：

将总结表文件重命名为index.htm，固定保存为htm格式，可使用Word等文字编辑软件进行内容编辑；

作品命名格式为：参赛单位+第一联系人姓氏+参赛赛道，示例：马里兰大学琼斯博士团队参赛MC2赛道

命名为UMD-Jones-MC2，多份作品提交可在名称末尾添加数字区分；

填写全体参赛队员姓名、所属单位、电子邮箱，指定一名队员作为第一联系人，负责对接赛事所有沟通问询工作；

标注队伍是否为学生队伍，学生队伍定义为由学生牵头、全员为学生组成的参赛队伍，课程项目类参赛作品均归为学生队伍作品；

罗列本次解题过程中使用的所有数据分析工具，Tableau、SAS等商用工具直接填写名称，非商用工具需标注工具名、开发主体以及相关介绍链接，队伍自研工具需写明开发背景与用途；

填写全队完成该份作品预估总投入工时，多份作品需单独填写对应工时；

勾选是否同意将参赛作品公开收录至可视分析标准案例库；

提交配备语音解说的演示视频，视频时长不得超过4分钟，格式统一为wmv。文件体积合规可直接打包提交，体积过大可上传至外网并附上有效访问链接，需保证赛事全程链接可正常访问。若无外网存放渠道，可发送邮件至指定邮箱沟通其他提交方式，同时需提交视频电子版用于存档。

视频制作参考要点：清晰说明所用可视分析软件的数据分析作用、数据分析切入点及选择依据、核心可视化设计思路、可视化图表与交互功能的实际分析价值；说明数据筛选与数据转换所用工具及执行顺序；阐述可视化呈现的异常数据在分析工作中的运用；精准定义文中专业术语；列明研究过程中设定的重要前提假设；完整作答对应赛道的所有赛题问题。

Packaging and Submitting Your Entry

原文

Rename the summary form to “index.htm” and save it on your local computer. Use this form to provide your answers either by adding text to the form itself or by linking to the separate files you need to provide.

Note: GitHub based entries must also provide a PDF of their entry form

Create a folder on your computer with the same name as your entry name. Save the completed answer form (index.htm), your design image, your optional video and storyboards, and your other images in this folder as well. The index.htm file should contain link to all your materials. Use relative links so they will still work when your materials are moved to another folder.

Zip the entire directory and save it in a file using the same entry name, for example, UMD-Jones-MC2.zip. If your zip file is greater than 50 MB in size, please post your video or other large materials on the internet instead of including it in your zip file. The submission system can only accept files smaller than 50 MB. > Note: Entries must be packaged using ZIP. Other formats such as rar and tar are not acceptable.

IEEE VAST Challenge 2026 uses the Precision Conference System (PCS) to handle the submission and reviewing process. PCS is available at <https://new.precisionconference.com/~vgtc/>. If you do not already have a login for the system, you must register first. This can be done using the links on the sign in page. Once you are logged into your account, please click the “Submissions” button, and then select Society: VGTC, Conference/Journal: VIS 2026, and Track: VIS 2026 VAST Challenge.

翻译

将总结表命名为index.htm保存至本地，可直接在表格内填写答案，也可在表格内插入外部文件链接完成作答。通过GitHub提交作品的参赛者，还需额外提交报名表PDF版本。

在本地新建与作品同名文件夹，将填写完成的indexhtm表格、设计图纸、演示视频、脚本草图及其他相关素材全部存入文件夹内，在index.htm中使用相对路径关联所有参赛材料，确保文件移动后链接依旧有效。

将整个文件夹压缩为ZIP格式压缩包，压缩包名称与作品名称保持一致。提交系统仅接收50MB以内的文件，压缩包超出大小限制时，需将视频等大体量素材外链上传，禁止放入压缩包内。赛事仅认可ZIP压缩格式，RAR、TAR等格式一律不予接收。

2026年IEEE VAST挑战赛统一使用PCS会议投稿系统完成作品提交与评审工作，投稿系统网址已给出。无账号参赛者需先完成注册，登录后点击提交入口，依次选择对应分类：协会VGTC、会议VIS 2026、赛事赛道VIS 2026 VAST Challenge即可完成投稿。

Optional Two-Page Summaries

原文

Participants may publish two page summaries of their submission in the Conference Proceedings. These summaries allow the contestant to give a general overview of their approach and tools, highlight novel features, provide references to papers and other relevant work and describe any new discoveries made while working through the Challenge problem.

The two-page summary must be formatted according to the IEEE VGTC Guidelines. They are submitted separately from your entry. They are due on August 14, 2026. (Please note revised date: this is required for inclusion in the IEEE conference proceedings.)

翻译

参赛者可自愿撰写两页精简论文，收录至赛事会议论文集。精简论文可概述解题思路、所用工具、创新设计点，引用相关学术文献，阐述参赛过程中取得的全新研究发现。

两页精简论文必须严格遵循IEEE VGTC排版规范，与参赛作品分开独立提交，提交截止时间为2026年8月14日，该文件是作品录入IEEE会议论文集的必要材料。